



APPLIED AI | NLP | DOCUMENT AUTOMATION

Bruno Ferreira

Applied AI Engineer / AI Solutions Engineer

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PROFILE Applied AI Engineer who turns document-heavy operational work into testable, evidence-guided software. I combine NLP, language models, information extraction and full-stack engineering, from process discovery and requirements to prototyping, evaluation, human validation and deployment preparation. My experience spans client delivery, research, university teaching and practical AI training. Based in Portugal and open to roles across the European Union.

Applied AI | NLP and language models | Document automation | AI evaluation | Full-stack delivery

01 / PROFESSIONAL EXPERIENCE

Independent Applied AI Consultant and Trainer

June 2025 - Present

Selected client work | Portugal and remote

- Take direct responsibility for process discovery, requirements, solution architecture, implementation, testing and technical documentation within each agreed project scope.
- Define evaluation and acceptance criteria, build in human-review checkpoints, prepare deployment handovers and deliver practical AI training to client teams.

Invited Lecturer

February 2024 - Present

University of Coimbra

- Modules taught: Principles of Databases, Fundamentals of Artificial Intelligence, Principles of Java, Principles of C, Natural Language Processing and Knowledge Representation.
- Preparation of learning materials, guidance and assessment of student work, and supervision of MSc dissertations.

Researcher

February 2022 - June 2025

CISUC | Centre for Responsible AI

- Conducted research in NLP, NLG, Transformers, Information Extraction, Question Answering, language models, Explainable AI, causal inference and model compression.
- Published and presented work on Question Answering, information extraction and the evaluation of knowledge encoded in language models.

02 / SELECTED APPLIED AI PROJECTS

AI platform for US visa workflows

2025 - Present

CONTEXT: Multi-tenant LegalTech platform that applies USCIS criteria to scattered documentation in a consistent, evidence-guided workflow. **MY CONTRIBUTION:** Designed and implemented workflow components for structured intake, triage, dynamic questionnaires, document management, OCR, evidence-guided language-model analysis and legal-review reports within a multi-tenant full-stack architecture.

TOOLS: Python, TypeScript, FastAPI, Next.js, PostgreSQL, Redis, Alembic, Docker, OCR, LLMs, ClamAV, Sentry

HVAC technical-specification extraction

2026

CONTEXT: Pipeline that transforms Portuguese HVAC specification PDFs into structured Excel and JSON outputs for engineering review. **MY CONTRIBUTION:** Designed and implemented a schema-driven hybrid extraction workflow combining embedding-based RAG, deterministic table mapping and regular expressions with vision and language models, preserving source-page evidence and benchmarking against manual ground truth.

TOOLS: Python, RAG, embeddings, regular expressions, PDF and table extraction, vision, LLMs, Excel, JSON, XAI

Quality Copilot for document validation

2026

CONTEXT: Specialised modules that support comparison and validation of packaging and technical documentation. **MY CONTRIBUTION:** Designed and built workflows for dimensional checks, content validation, correction verification and technical-sheet preparation, defining evidence requirements and human-review gates for every stage.

TOOLS: Python, OCR, Claude Code skills, document automation

Global THE 17.3 evidence evaluation

2025

CONTEXT: Commissioned to design and carry out a systematic review of public university information for THE Impact Rankings indicators 17.3.1 to 17.3.17. **MY CONTRIBUTION:** Defined the evidence-review method and a repeatable assessment process, applied consistently across the 17 indicators.

TOOLS: Evidence review, evaluation design, structured analysis

Privacy-by-design AI platform

Product in development

CONTEXT: Own product combining local personal-data detection and anonymisation with external language-model workflows. **MY CONTRIBUTION:** Defined the product strategy and architecture and built a functional PWA with document processing, model-assisted chat, projects, user profiles and verifiable privacy controls.

TOOLS: React, TypeScript, NER, ONNX, LLMs, Privacy by Design

03 / TECHNICAL CAPABILITIES

Applied AI: NLP, language models, Information Extraction, Question Answering, OCR, XAI, prompting, evaluation and human review

Engineering: Python, TypeScript, React, Next.js, FastAPI, SQL, PostgreSQL, APIs, Git, Unix, Docker and document automation

From problem to delivery: Needs discovery, prototyping, testing, technical review, documentation, deployment preparation and training

Development with AI tools: Claude Code, Codex, OpenCode and Pi for implementation, review, testing and documentation, with defined review criteria and my own supervision

04 / EDUCATION

PhD in Informatics Engineering

2022 - Present

University of Coimbra

MSc in Informatics Engineering - Intelligent Systems

2022

University of Coimbra

BSc in Informatics Engineering

2021

University of Coimbra

05 / SELECTED PUBLICATIONS

- Towards Automated Evaluation of Knowledge Encoded in Large Language Models | LREC-COLING, 2024
- BATS-PT: Assessing Portuguese Masked Language Models | PROPOR, 2024
- Leveraging Question Answering for Domain-Agnostic Information Extraction | CIARP, 2023
- Evaluating the Extraction of Toxicological Properties with Extractive Question Answering | EANN, 2023
- SafetyDesk: Accelerating the Production of Toxicological Reports | EXP.AT, 2023
- ETFs Information Extraction from KIIDs | Environmental Finance for the Common Good, 2023
- Question Answering for Toxicological Information Extraction | SLATE, 2022
- Safety Desk: Extraction and Analysis of Textual Information to Build a Reporting System | MSc dissertation, 2022

06 / LANGUAGES

Portuguese - Native | English - Advanced | Spanish - Good comprehension, basic speaking

07 / PROFESSIONAL LINKS

[LinkedIn](#) | [Portfolio](#) | [Ciência Vitae](#) | [ORCID](#)